



Installation and Commissioning Quality Control Record

Short circuit saturation test

Q-1523-010

Revision # 00

Page 1 of 2

ITP step #: 1520-15

Project: BCH GMS

Unit: 4

Rated Armature Amps 13430 amps

% Armature Amps	Armature Amps	Line 1	Line 2	Line 3	Field Volts	Field Amps	Frequency Hz
109.8	14739.5	14730	14775	14704	158.9	1241.3	59.9
100.6	13515.5	13507	13546	13485	147.5	1136.2	59.9
90.4	12134.5	12129	12160	12109	128.4	1014.2	60.0
80.5	10812.5	10808	10835	10790	117.5	905.4	60.2
70.5	9469.0	9466	9488	9450	104.5	789.7	60.2
60.1	8071.5	8068	8087	8056	86	673.3	60.2
50.0	6714.0	6713	6727	6701	73.9	556.4	60.1
40.1	5379.0	5377	5390	5368	60.7	448.4	60.2
30.1	4044.5	4039	4049	4040	43.8	334.1	60.1
20.3	2722.0	2707	2714	2730	29.8	222.0	60.2
10.2	1365.5	1360	1363	1368	15.8	109.9	60.2
0.6	84.8	84.2	84.6	85	0	2.9	60.0

- * Readings taken in "Descending" order
- * If a point is missed do not go back - hysteresis effect will distort reading - continue or start over !!!
- * Point "0" - Magnetization/Residual voltage at zero excitation & field breaker closed
- * Frequency/Hz to remain constant during readings
- * Ensure line current balance between phases

Measurement Instruments	T°	Measured by: Winston Crawford	05/02/2013
Yokogawa 2030	Before: 20		
HCP 0035	After: 20	Andritz Rep.: Jean-Luc Cote	
Fluke 189 - HCP 0184		Cust. Repres.: Yutao Zhang	



Installation and Commissioning
Quality Control Record

Short circuit saturation test

Q-1523-010

ITP step #: 1520-15

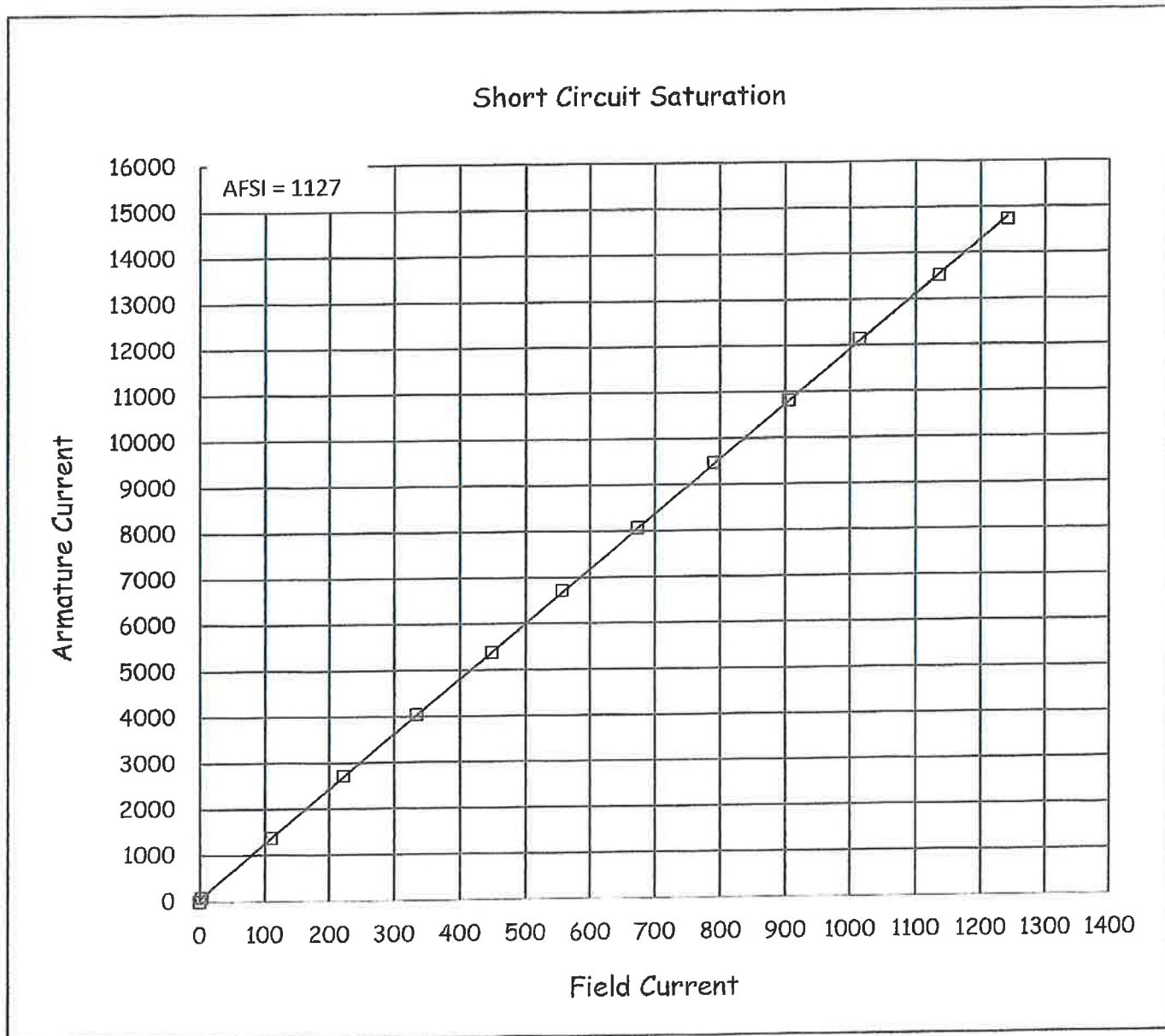
Revision # 00

Project: BCH GMS

Unit : 4

Page 2 of 2

Short circuit saturation curve



Measurement Instruments

Yokogawa 2030

HCP 0035

Fluke 189 - HCP 0184

T°

Before: 20

After: 20

Measured by: Winston Crawford

05/02/2013

ANDRITZ Rep.: Jean-Luc Cote

Cust. Repres.: Yutao Zhang

Signature: [Handwritten Signature] Date: 2/10/2013



Installation and Commissioning
Quality Control Record

Open circuit saturation test

Q-1523-020

ITP step #: 1520-17
Project: BCH GMS

Unit : 4

Revision # 01
Page 1 of 3

Rated Armature Volts 13800 Volts

% Armature Volts	Armature Volts	Line 1	Line 2	Line 3	Field Volts	Field Amps	Frequency Hz
		171	171	171	0.0	0.0	60.13
1.2	171	1408	1408	1408	10.8	79.8	60.14
10.2	1408	2746	2746	2747	19.5	156.4	60.15
19.9	2746	4129	4129	4129	29.0	236.9	60.13
29.9	4129	5540	5539	5540	39.8	316.8	60.13
40.1	5540	6903	6902	6903	46.8	395.6	60.13
50.0	6903	8288	8287	8288	59.8	477.5	60.12
60.1	8288	8979	8978	8979	65.2	521.9	60.15
65.1	8979	9669	9668	9669	70.9	566.6	60.13
70.1	9669	10352	10352	10352	77.1	613.2	60.13
75.0	10352	11056	11054	11055	83.8	667.2	60.13
80.1	11055	11749	11748	11748	90.9	723.0	60.14
85.1	11748	12444	12443	12444	98.8	784.4	60.18
90.2	12444	13130	13128	13129	108.2	857.9	60.08
95.1	13129	13822	13821	13821	118.2	934.9	60.15
100.2	13821	14515	14513	14514	130.4	1032.3	60.14
105.2	14514	15194	15193	15193	145.0	1147.1	60.11
110.1	15193	15894	15892	15893	162.8	1284.6	60.13
115.2	15893	16583	16581	16582	188.7	1486.1	60.00
120.2	16582						

- * Readings taken in "Ascending" order
- * Point "0" - Magnetization/Residual voltage at zero excitation & field breaker closed
- * Frequency/Hz to remain constant during readings
- * Ensure line voltage balance between phases

NCR #: _____		Measured by: <u>Winston Crawford</u>		08/02/2013
Measurement Instruments	T°			
Yokogawa 2030	Before: <u>20</u>			
HCP 0035	After: <u>20</u>			
Fluke 189 - HCP 0184		ANDRITZ Rep.: <u>Jean-Luc Cote</u>		
		Cust. Repres.: <u>Yutao Zhang</u>		

Installation and Commissioning

Quality Control Record

Open circuit saturation test

Q-1523-020

ITP step #: 1520 - 17

Revision # 01

Project: BC Hydro GMS

Unit : 4

Page 2 of 3

[illegible]

Vibrations readings are looked after in the mechanical run (ITP step 1520-10 QCR Q-1522-010) and as such this page is not applicable.

NCR #: _____

Measurement Instruments _____ T° _____ Before: _____ _____ After: _____ _____ _____	Measured by: _____ dd-mmm-yyyy ANDRITZ Rep.: <u>Kristin Perles & [Signature]</u> Cust. Repres.: _____
---	---

Open circuit saturation test

Q-1523-020

ITP step #: 1520-17

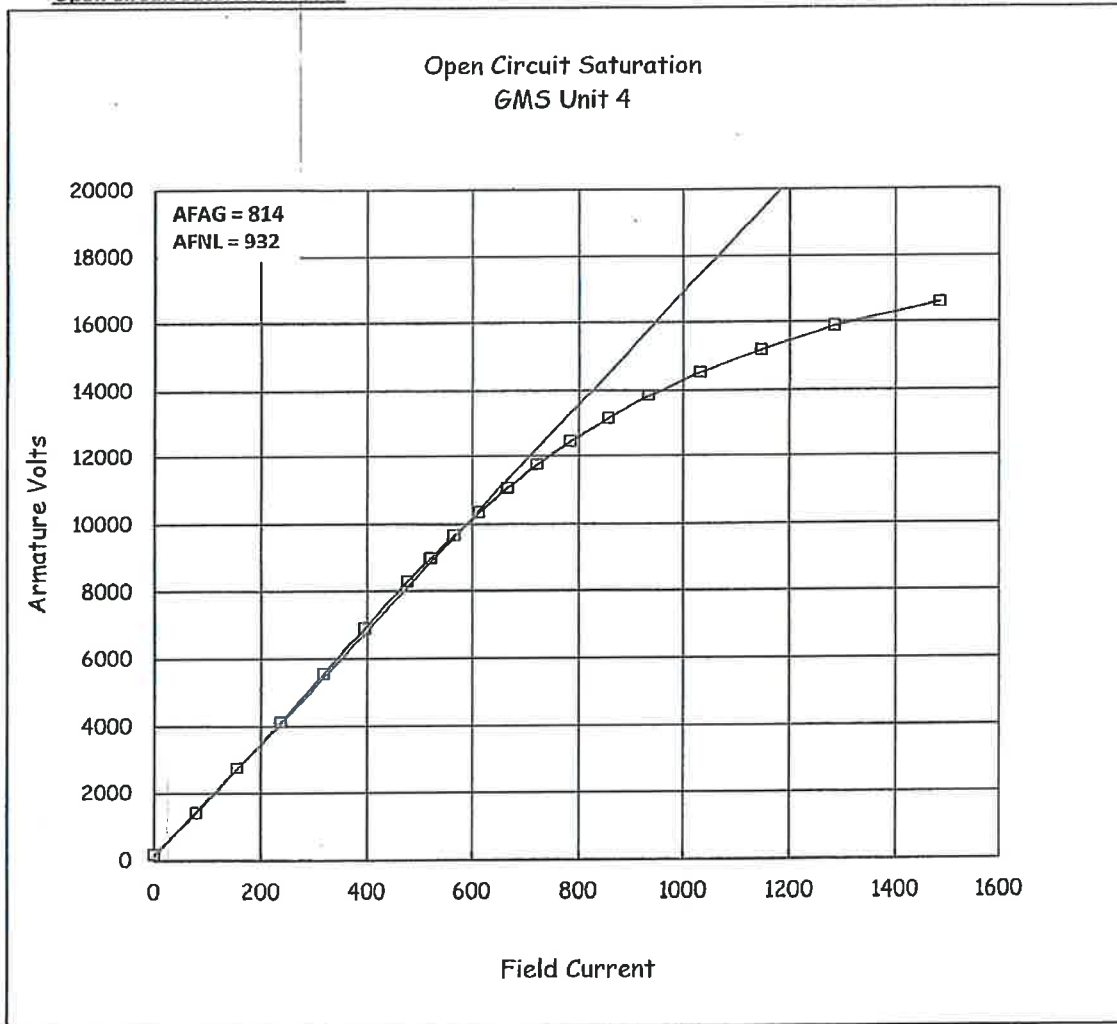
Revision # 01

Project: BCH GMS

Unit : 4

Page 3 of 3

Open circuit saturation curve



NCR #: _____

Measurement Instruments	T°	Measured by: <u>Winston Crawford</u>	08/02/2013
<u>Yokogawa 2030</u>	Before: <u>20</u>		
<u>HCP 0035</u>	After: <u>20</u>	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	
<u>Fluke 189 - HCP 0184</u>		Cust. Repres.: <u>Yutao Zhang</u>	



Installation and Commissioning
Quality Control Record

Waveform, Harmonics and TIF Tests

Q-1543-010

ITP step #: 1520-18

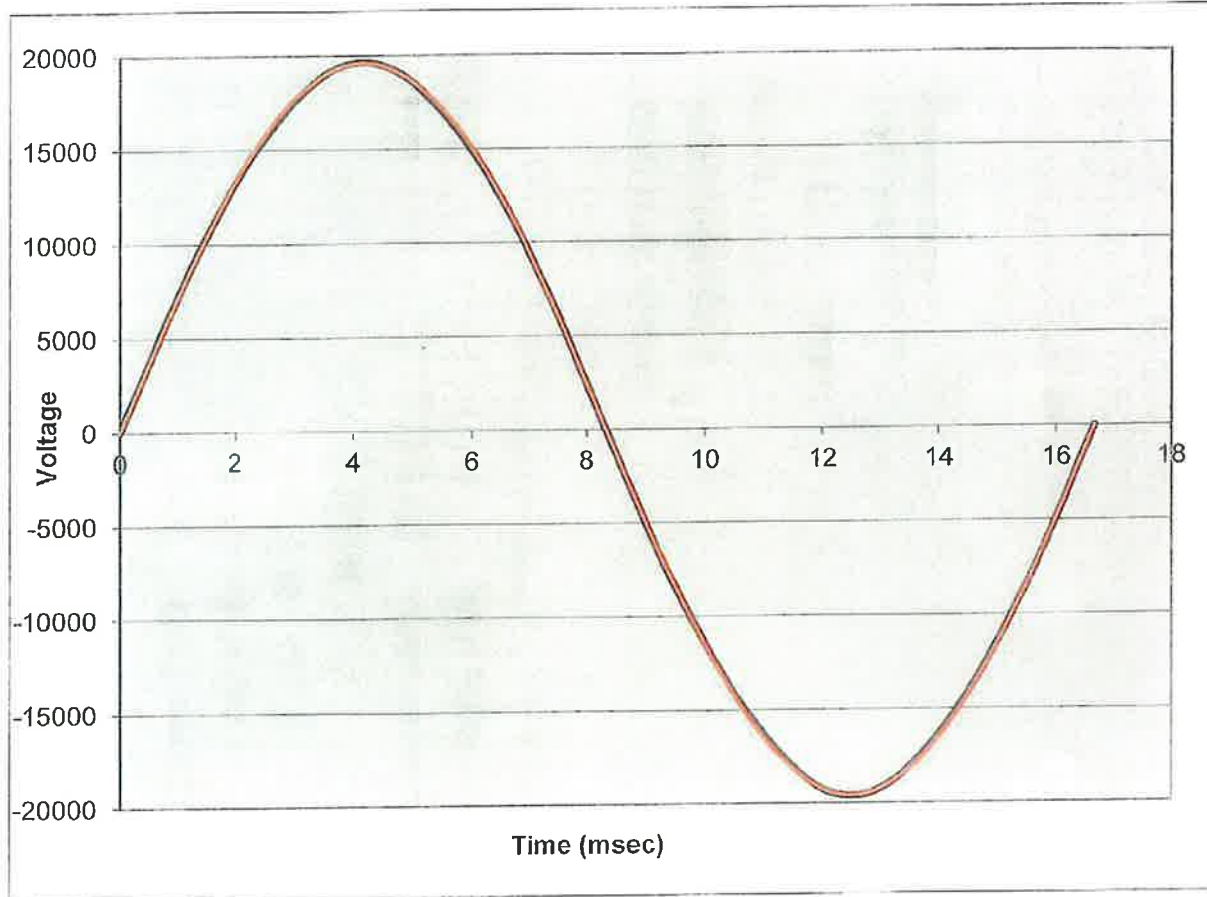
Project: BCH GMS

Unit: 4

Revision # 00

Page 1 of 9

Waveform Line - Line



Test voltage (shown in red) is plotted overtop of a pure sine wave (underneath) in black.

Average Armature Voltage Measured Line-to-Line =
Frequency at Test

13834
60

RMS

NCR #: _____

Measurement Instruments	T°	Measured by: Winston Crawford	08/02/2013
Dactron Signal Analyser	Before: 20		
SN # - 7213146	After: 20	ANDRITZ Rep.: Jean-Luc Cote	
		Cust. Repres.: Yutao Zhang	



Installation and Commissioning
Quality Control Record

Waveform, Harmonics and TIF Tests

Q-1543-010

ITP step #: 1520-18

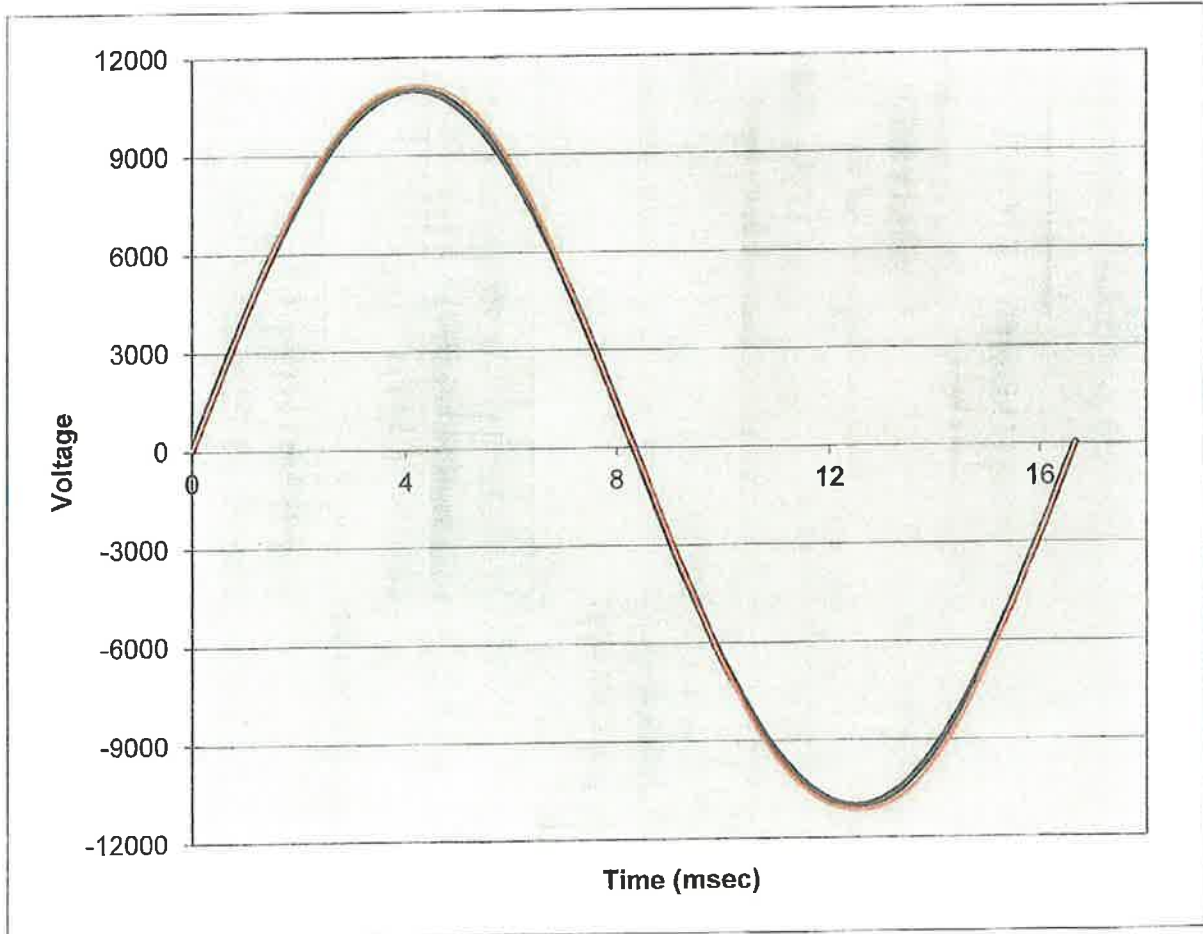
Project: BCH GMS

Unit : 4

Revision # 00

Page 2 of 9

Waveform Line - Neutral



Test voltage (shown in red) is plotted overtop of a Pure sine wave (underneath) in black.

Average Armature Voltage Measured Line A to Neutral = 6922 RMS

Frequency at test = 60

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Winston Crawford</u>	<u>08/02/2013</u>
Dactron Signal Analyser	Before: _____		
SN # - 7213146	After: _____	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	
		Cust. Repres.: <u>Yutao Zhang</u>	<u>3/4/13</u>



Installation and Commissioning
Quality Control Record

Waveform, Harmonics and TIF Tests

Q-1543-010

ITP step #: 1520-18

Project: BCH GMS

Unit : 4

Revision # 00

Page 3 of 9

Deviation Factors

Rated Armature Voltage Line-to-Line (Peak)				19516
Maximum absolute L-L voltage recorded during test period				19619
Rated Armature Voltage L-N (Peak)				11268
Maximum absolute L-N voltage recorded during test period				11138
Hertz	Line-Line	Percent	Line-Neutral	Percent
	Vac RMS	of	Vac RMS	of
	C-A	Fundamental	A - N	Fundamental
60	13834	100.25	6922	87

The RMS values of the waveforms were determined using the acquired data.

Deviation Factor Summary				
	LA - LB		LA - N	
Maximum Deviation volts				
	260.3	volts	290.3	volts
Maximum Deviation volts / Peak volts				
	0.01		0.03	
Deviation Factor				
	1.33%		2.63%	

See Graphs for a comparison of the test waveforms to sine waves that were calculated based on the acquired data.

NCR #: _____

Measurement Instruments	T°	Measured by: Winston Crawford	08/02/2013
Dactron Signal Analyser	Before: 20		
SN # - 7213146	After: 20	ANDRITZ Rep.: Jean-Luc Cote	
		Cust. Repres.: Yutao Zhang	



Installation and Commissioning
Quality Control Record

Waveform, Harmonics and TIF Tests

Q-1543-010

ITP step #: 1520-18

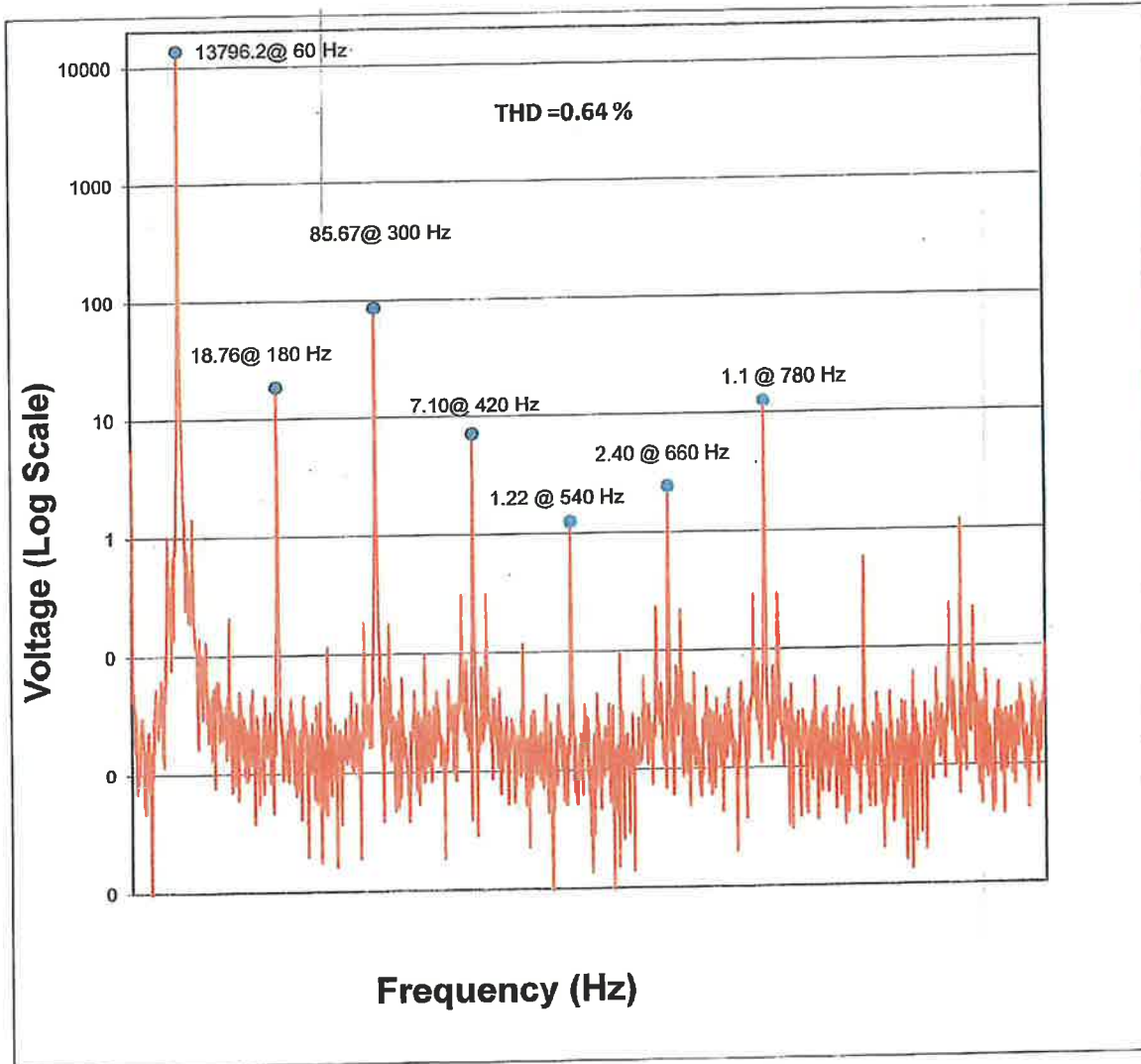
Project: BCH GMS

Unit : 4

Revision # 00

Page 4 of 9

Spectrum Line-To-Line



NCR #: _____

Measurement Instruments	T°	Measured by: Winston Crawford	08/02/2013
Dactron Signal Analyser	Before: 20		
SN # - 7213146	After: 20	ANDRITZ Rep.: Jean-Luc Cote	
		Cust. Repres.: Yutao Zhang	



Installation and Commissioning
Quality Control Record

Waveform, Harmonics and TIF Tests

Q-1543-010

ITP step #: 1520-18

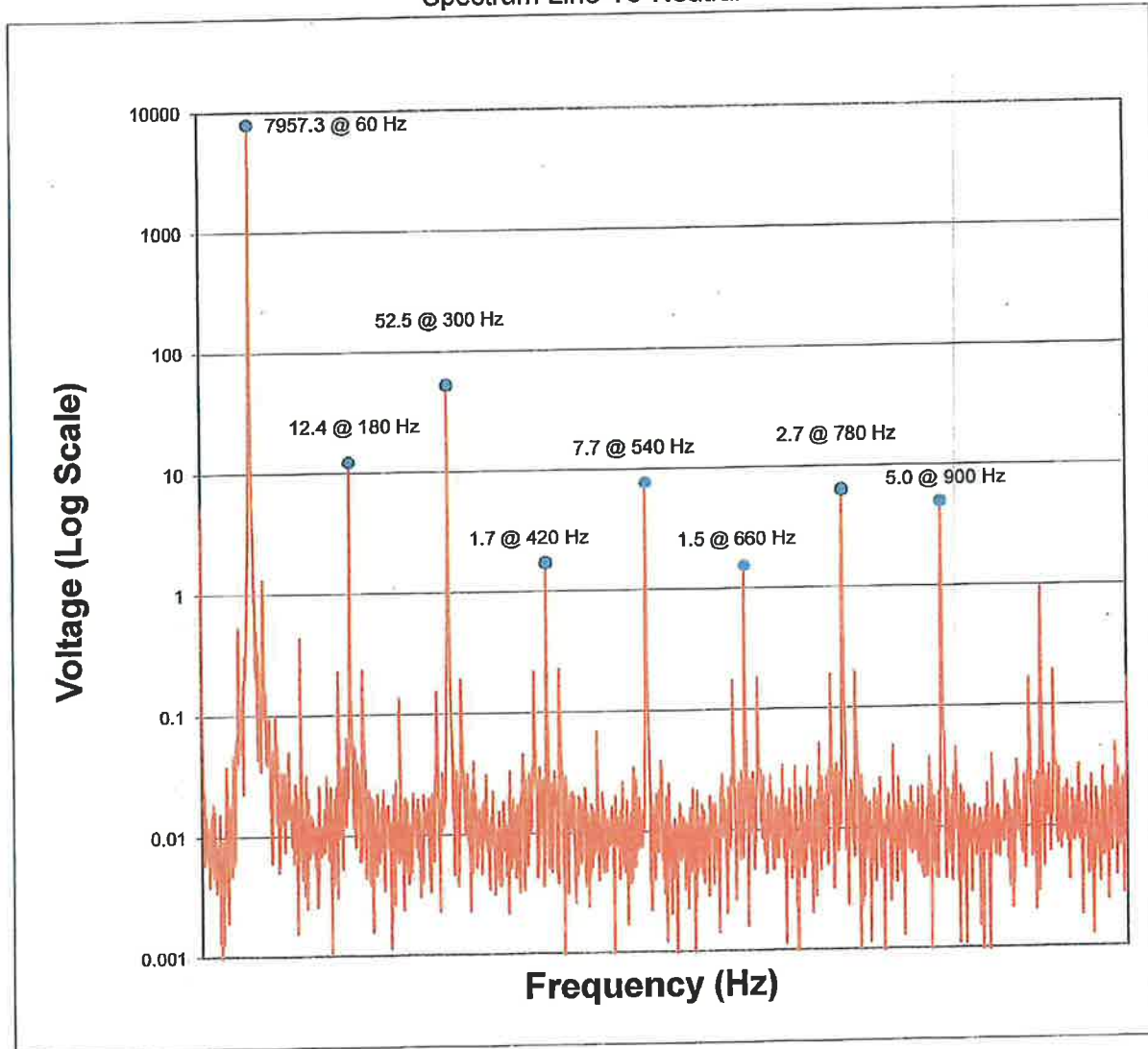
Project: BCH GMS

Unit : 4

Revision # 00

Page 5 of 9

Spectrum Line-To-Neutral



NCR #: _____

Measurement Instruments	T°	Measured by:	Winston Crawford	08/02/2013
Dactron Signal Analyser	Before: 20			
SN # - 7213146	After: 20	ANDRITZ Rep.:	Jean-Luc Cote	
		Cust. Repres.:	Yutao Zhang	



Installation and Commissioning
Quality Control Record

Waveform, Harmonics and TIF Tests

Q-1543-010

ITP step #: 1520-18

Project: BCH GMS

Unit : 4

Revision # 00

Page 6 of 9

TIFF Balanced

Hertz	TIF F	Line-To-Line Vac RMS	Tif f * Measure	Calculated Squared
60	0.5	13796.24	6898.12	47584075
180	30	18.76	562.67	316602
300	225	85.67	19274.91	371522096
360	400	0.10	40.27	1621
420	650	7.10	4612.76	21277509
540	1320	1.33	1758.58	3092597
660	2260	2.41	5441.30	29607731
720	2760	0.02	60.35	3642
780	3360	8.37	28109.83	790162423
900	4350	0.43	1871.93	3504105
1000	5000	0.01	38.79	1505
1020	5100	1.17	5974.04	35689111
1080	5400	0.02	95.63	9146
1140	5630	1.24	7002.23	49031194
1260	6050	0.21	1288.74	1660847
1380	6370	0.80	5071.42	25719276
1440	6650	0.05	363.39	132049
1500	6680	2.06	13746.38	188963004
1620	6970	0.16	1148.33	1318654
1740	7320	0.83	6053.36	36643126
1800	7570	0.05	349.77	122339
1860	7820	1.74	13568.02	184091112
1980	8330	0.07	583.96	341013
2100	8830	0.85	7540.65	56861335
2160	9080	0.03	259.74	67464
2220	9330	0.69	6438.07	41448786
2340	9840	0.14	1365.77	1865332

Measurement Instruments	T°	Measured by:	Winston Crawford	08/02/2013
Dactron Signal Analyser	Before: <u>20</u>			
SN # - 7213146	After: <u>20</u>	ANDRITZ Rep.:	Jean-Luc Cote	
		Cust. Repres.:	Yutao Zhang	



Installation and Commissioning
Quality Control Record

Waveform, Harmonics and TIF Tests

Q-1543-010

ITP step #: 1520-18

Project: BCH GMS

Unit : 4

Revision # 00

Page 7 of 9

TIFF Balanced

Hertz	TIF F	Line-To-Line Vac RMS	Tif f * Measure	Calculated Squared
2160	9080	0.03	259.74	67464
2220	9330	0.69	6438.07	41448786
2340	9840	0.14	1365.77	1865332
2460	10340	0.47	4844.37	23467968
2580	10600	0.63	6635.11	44024629
2820	10210	0.44	4487.54	20138040
2940	9820	0.38	3731.47	13923834
3000	9670	0.06	553.87	306769
3180	8740	0.28	2451.83	6011493
3300	8090	0.27	2169.68	4707501
3540	6730	0.18	1181.75	1396531
3660	6130	0.23	1429.00	2042038
3900	4400	0.17	760.86	578902
4020	3700	0.13	468.81	219781
4260	2750	0.05	150.39	22618
4380	2190	0.09	189.09	35754
5000	840	0.01	9.76	95

Sum	2007913648.7
Sq. Root	44809.7
(Balanced) L-L tif	3.2

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Winston Crawford</u>	08/02/2013
Dactron Signal Analyser	Before: <u>20</u>		
SN # - 7213146	After: <u>20</u>	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	
		Cust. Repres.: <u>Yutao Zhang</u>	



Installation and Commissioning
Quality Control Record

Waveform, Harmonics and TIF Tests

Q-1543-010

ITP step #: 1520-18

Project: BCH GMS

Unit : 4

Revision # 00

Page 8 of 9

TIFF Line-To-Neutral

Hertz	TIF F	Line-To-Neutral Vac RMS	Tif f * Measure	Calculated Squared
60	0.5	7957.32	3978.66	15829727
180	30	12.41	372.34	138635
300	225	52.57	11827.78	139896471
360	400	0.00	1.72	3
420	650	1.73	1121.46	1257666
540	1320	7.73	10206.50	104172728
660	2260	1.55	3497.63	12233398
720	2760	0.03	88.42	7819
780	3360	6.41	21548.44	464335464
900	4350	4.06	17640.38	311182850
1000	5000	0.00	23.40	548
1020	5100	0.95	4868.44	23701691
1080	5400	0.03	136.66	18675
1140	5630	1.40	7907.68	62531324
1260	6050	0.00	0.00	0
1380	6370	0.95	6071.99	36869059
1440	6650	0.00	0.00	0
1500	6680	3.10	20696.93	428362989
1620	6970	0.04	279.53	78136
1740	7320	0.85	6215.12	38627763
1800	7570	0.02	164.00	26896
1860	7820	0.94	7319.58	53576285
1980	8330	0.01	117.58	13825
2100	8830	0.68	5981.77	35781573
2160	9080	0.01	107.58	11574
2220	9330	0.75	6957.86	48411851
2340	9840	0.05	464.82	216060

NCR #:

Measurement Instruments	T°	Measured by:	Winston Crawford	08/02/2013
Dactron Signal Analyser	Before: 20			
SN # - 7213146	After: 20	ANDRITZ Rep.:	Jean-Luc Cote	
		Cust. Repres.:	Yutao Zhang	



Installation and Commissioning
Quality Control Record

Waveform, Harmonics and TIF Tests

Q-1543-010

ITP step #: 1520-18

Project: BCH GMS

Unit: 4

Revision # 00

Page 9 of 9

TIFF Line-To-Neutral

Hertz	TIF F	Line-To-Neutral Vac RMS	Tif f * Measure	Calculated Squared
2460	10340	0.57	5926.95	35128773
2580	10600	0.50	5328.94	28397644
2820	10210	0.46	4658.52	21701851
2940	9820	0.39	3842.28	14763109
3000	9670	0.01	126.14	15912
3180	8740	0.43	3799.56	14436635
3300	8090	0.38	3074.55	9452888
3540	6730	0.36	2429.01	5900113
3660	6130	0.35	2120.26	4495523
3900	4400	0.28	1235.27	1525889
4020	3700	0.26	967.12	935314
4260	2750	0.21	572.19	327401
4380	2190	0.19	418.45	175097
5000	840	0.01	4.30	18

Sum	1914539176.8
Sq. Root	43755.4
(Balanced) L-N tif	5.5

NCR #: _____

Measurement Instruments	T°	Measured by: Winston Crawford	08/02/2013
Dactron Signal Analyser	Before: 20		
SN # - 7213146	After: 20	ANDRITZ Rep.: Jean-Luc Cote	
		Cust. Repres.: Yutao Zhang	



Quality Control Record

Generator Heat Run

Q-1524-040

ITP Step # 1520-19

Project BC Hydro GMS

Unit : 4

Page 1 of 9

	Date	2013/02/15	2013/02/18	2013/02/19
	Time	10:45 - 11:45	07:00 - 08:00	07:00 - 08:00
	Heat run	1	2	3
Output MW		58.58	217.51	255.82
Output MVA		58.72	227.70	268.57
Output MVAR		0.79	67.59	81.62
Power Factor		0.9975	0.9550	0.9526
Hertz		60.03	60.00	60.03
Armature Voltage L-L		13993.6	14456.0	14585.3
Armature Current		2430.6	9096.9	10629.0
Field Voltage		139.64	237.60	267.43
Field Current		1032.47	1599.60	1749.50
Field I ² R		144.18	380.06	467.87
SAC Air Outlet Temp. (Avg)		27.8	28.1	28.6
Maximum Armature RTD		55.8	84.6	95.2
Average Armature RTD		54.2	81.7	91.7
Minimum Armature RTD		52.5	78.3	87.9
Armature Temperature Rise		28.0	56.5	66.7
Total Field Temperature		49.8	77.7	86.8
Field Temperature Rise		22.0	49.6	58.2

Shaft Runouts	Total - Ampl/Angle		
	(Mils P-P)/Deg.		
Gen'r Bearing "X"	7.9 - 2.5/116	1.6 - 0.4/126	1.4 - 0.4/111
Gen'r Bearing "Y"	6.6 - 2.3/18	2.4 - 1.3/7	2.4 - 1.3/5
Turbine Bearing "X"	4.6 - 0.3/na	2.2 - 1.1/315	2.4 - 1.5/326
Turbine Bearing "Y"	4.2 - 0.3/na	2.5 - 1.3/241	2.7 - 1.9/248

- Notes :
1. The stabilized readings were averaged
 2. Rated Armature Voltage is 13800 volts
 3. Rated Armature Current is amps 13430

Measured by:	<u>Winston Crawford</u>	Date : <u>20/02/2013</u>
ANDRITZ Rep.:	<u>Jean-Luc Cote</u>	Revised - <u>May 8th, 2013</u>
Cust. Repres.:	<u>Yutao Zhang</u>	By - <u>W.G. Crawford</u>



Quality Control Record

Generator Heat Run

Q-1524-040

ITP Step # 1520-19

Project BC Hydro GMS

Unit : 4

Page 2 of 9

	Heat Run 1			15-Feb
TIME	10:45	11:15	11:45	Average
Armature Voltage L-L - Average	13985.3	13998.3	13997.0	13993.6
Armature Current - Average	2411.7	2477.7	2402.3	2430.6
MVA Reading	58.43	60.07	57.67	58.72
MW Reading	58.24	59.96	57.54	58.58
P.F. Reading	0.9967	0.9981	0.9976	0.9975
MVARs Reading	-1.33	1.15	2.55	0.79
Field Voltage				
K = 1	138.30	140.58	140.05	139.64
Field Current				
K = 1	1022.85	1039.65	1034.92	1032.47
Field Ohms				
	0.1352104	0.1352186	0.1353245	0.1352512
Field Max Temperature				
	49.7	49.7	49.9	49.8
SAC Air Outlet Temperature (Average of 8 coolers)				
	28.0	27.8	27.5	27.8
Armature Winding RTD's (Avg.)				
Max	55.0	55.8	56.5	55.8
Min	51.8	52.5	53.1	52.5
Avg	53.5	54.3	54.9	54.2

Measured by: Winston Crawford

Date :20/02/2013

ANDRITZ Rep.: Jean-Luc Cote

Revised - May 8th, 2013

Cust. Repres.: Yutao Zhang

By - W.G. Crawford



Quality Control Record

Generator Heat Run

Q-1524-040

ITP Step # 1520-19

Project BC Hydro GMS

Unit : 4

Page 3 of 9

	Heat Run 2			18-Feb
TIME	07:00	07:30	08:00	Average
Armature Voltage L-L- Average	14451.3	14459.7	14457.0	14456.0
Armature Current - Average	9175.0	9129.7	8986.0	9096.9
MVA Reading	229.45	228.65	225.00	227.70
MW Reading	219.45	217.57	215.52	217.51
P.F. Reading	0.9556	0.9515	0.9579	0.9550
MVARs Reading	67.74	70.35	64.67	67.59
Field Voltage K = 1	237.50	238.25	237.05	237.60
Field Current K = 1	1598.47	1605.37	1594.95	1599.60
Field Ohms	0.1485796	0.1484082	0.1486253	0.1485377
Field Max Temperature	77.8	77.4	77.9	77.7
SAC Air Outlet (Average of 8 coolers)	28.1	28.0	28.1	28.1
Armature Winding RTD's				
Max	84.6	84.6	84.6	84.6
Min	78.2	78.2	78.4	78.3
Avg	81.7	81.6	81.7	81.7

Measured by: Winston Crawford

Date :20/02/2013

ANDRITZ Rep.: Jean-Luc Cote

Revised - May 8th, 2013

Cust. Repres.: Yutao Zhang

By - W.G. Crawford



Quality Control Record

Generator Heat Run

Q-1524-040

ITP Step # 1520-19

Project BC Hydro GMS

Unit : 4

Page 4 of 9

	Heat Run 3			19-Feb
TIME	07:00	07:30	08:00	Average
Armature Voltage L-L- Average	14588.0	14588.3	14579.7	14585.3
Armature Current - Average	10661.3	10709.7	10516.0	10629.0
MVA Reading	269.46	270.69	265.55	268.57
MW Reading	255.38	256.84	255.24	255.82
P.F. Reading	0.9477	0.9488	0.9612	0.9526
MVARs Reading	86.02	85.52	73.33	81.62
Field Voltage K = 1	269.32	270.06	262.91	267.43
Field Current K = 1	1763.62	1766.10	1718.77	1749.50
Field Ohms	0.152709	0.152913	0.152964	0.152862
Field Max Temperature	86.5	86.9	87.0	86.8
SAC Air Outlet (Average of 8 coolers)	28.5	28.6	28.6	28.6
Armature Winding RTD's				
Max	95.2	95.4	95.6	95.4
Min	87.5	88.0	88.1	87.9
Avg	91.4	91.6	92.0	91.7

Measured by: Winston Crawford

Date :20/02/2013

ANDRITZ Rep.: Jean-Luc Cote

Revised - May 8th, 2013

Cust. Repres.: Yutao Zhang

By - W.G. Crawford



Quality Control Record

Generator Heat Run

Q-1524-040

ITP Step # 1520-19

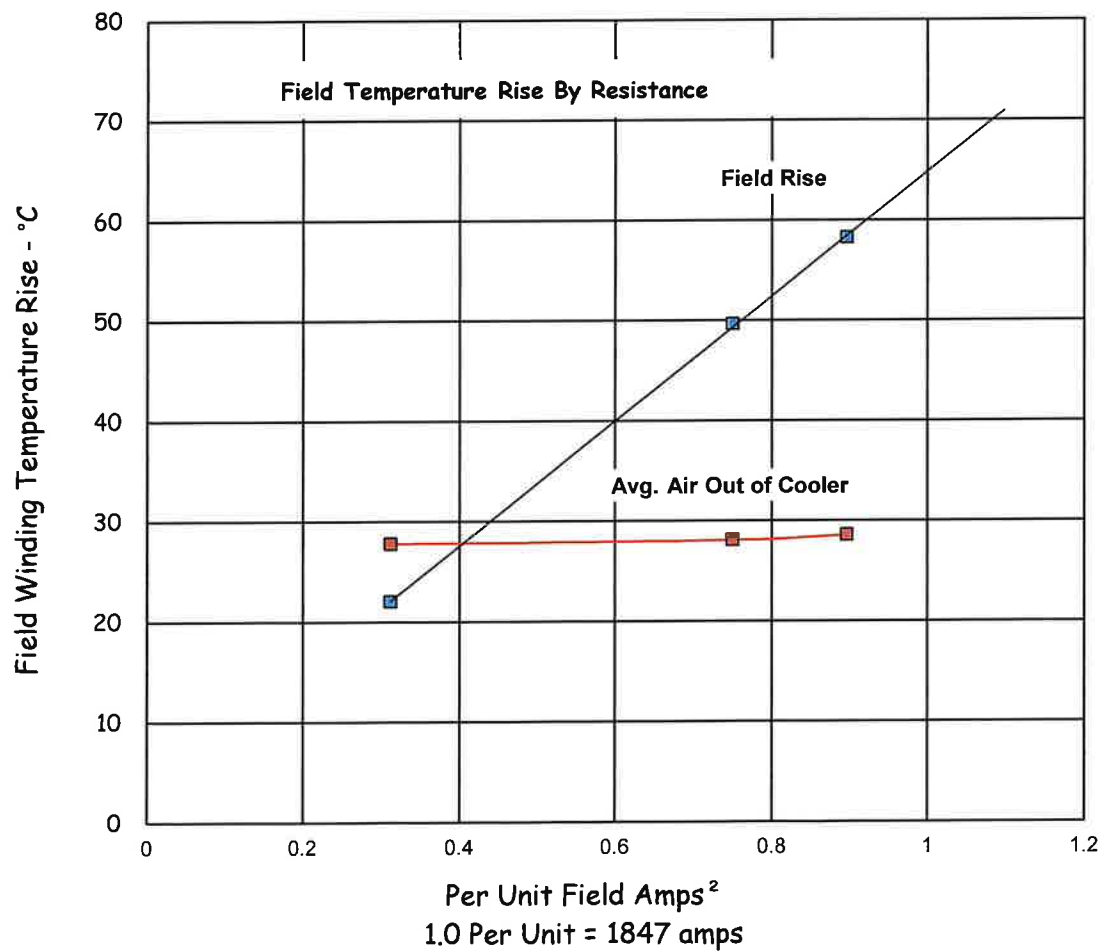
Project BC Hydro GMS

Unit : 4

Page 5 of 9

BCH GMS Unit 4

Heat Run Temperatures - Generator Field



Measured by: Winston Crawford

Date :20/02/2013

ANDRITZ Rep.: Jean-Luc Cote

Revised - May 8th, 2013

Cust. Repres.: Yutao Zhang

By - W.G. Crawford



Quality Control Record

Generator Heat Run

Q-1524-040

ITP Step # 1520-19

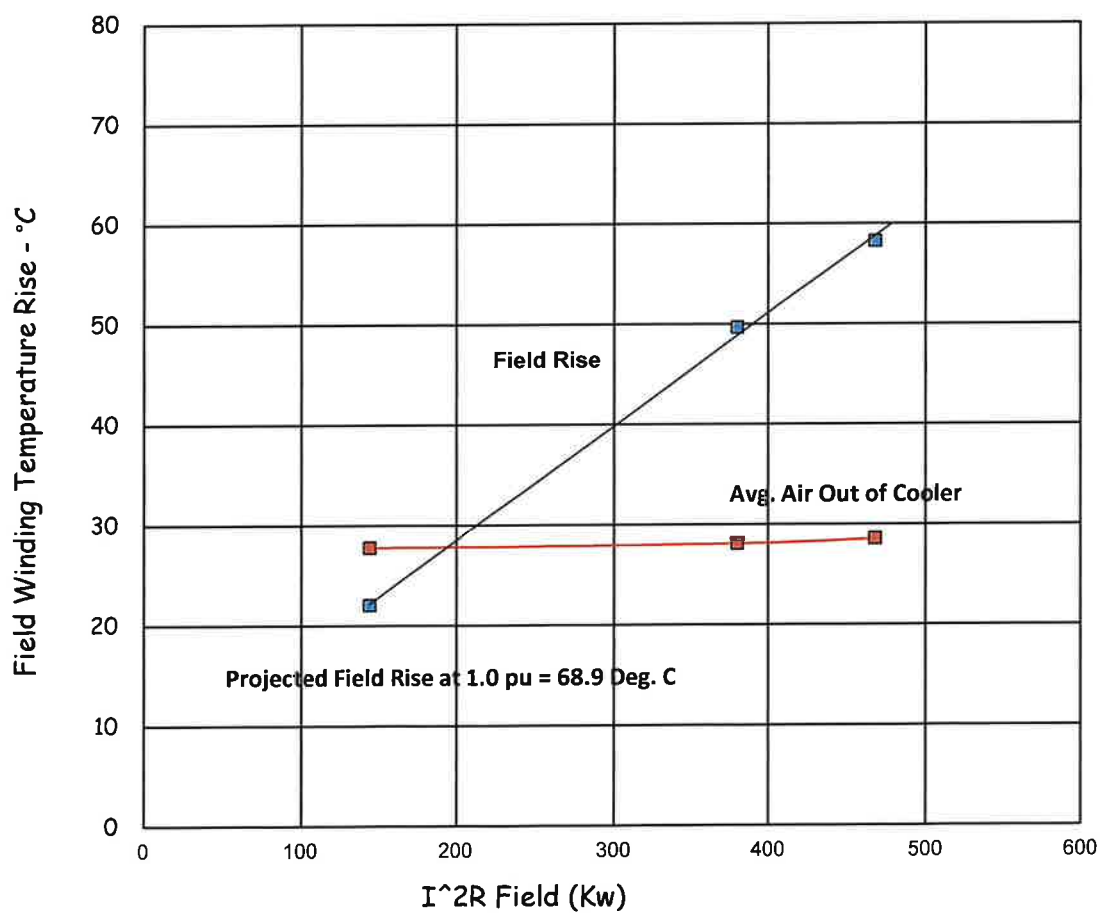
Project BC Hydro GMS

Unit : 4

Page 6 of 9

BCH GMS Unit 4

Heat Run Temperatures - Generator Field



Measured by: Winston Crawford

Date :20/02/2013

ANDRITZ Rep.: Jean-Luc Cote

Revised - May 8th, 2013

Cust. Repres.: Yutao Zhang

By - W.G. Crawford

ITP Step # 1520-19

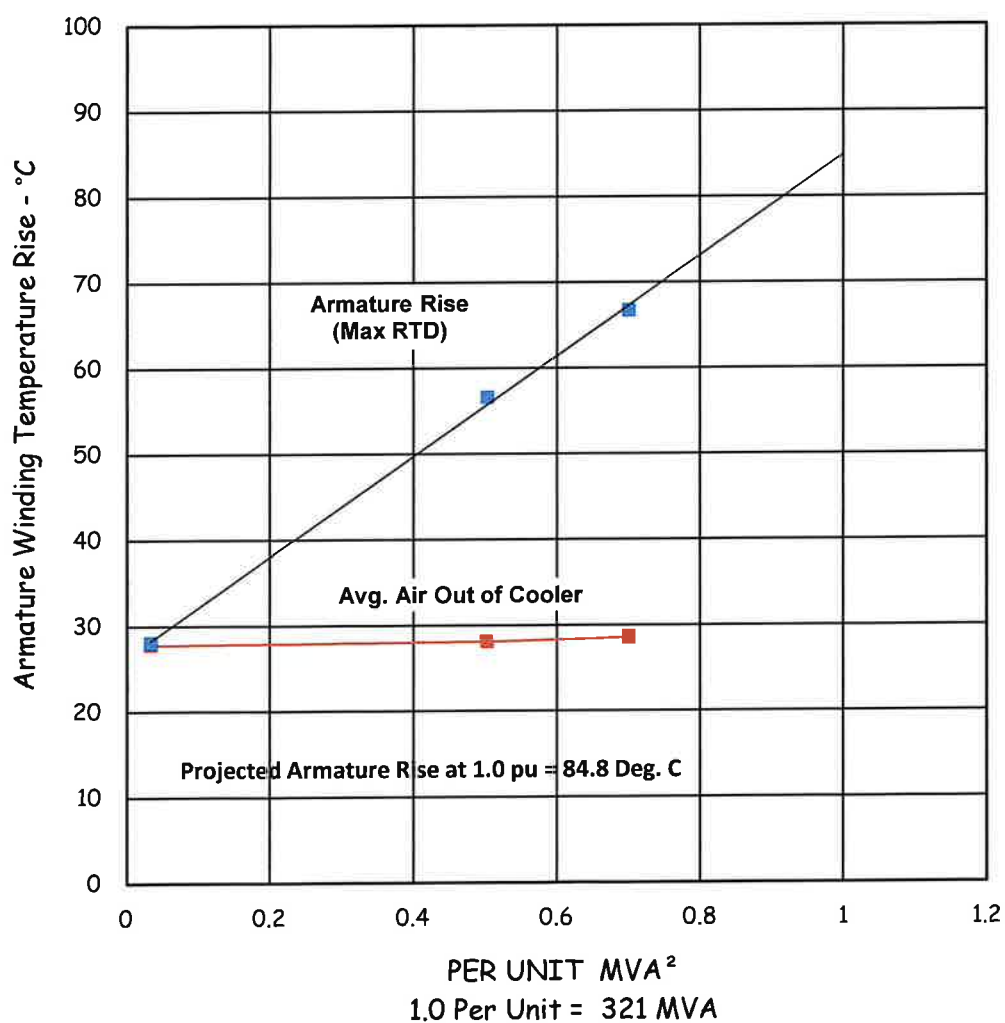
Project BC Hydro GMS

Unit : 4

Page 7 of 9

BCH GMS Unit 4

Heat Run Temperatures - Generator Armature



Measured by: Winston Crawford

Date :20/02/2013

ANDRITZ Rep.: Jean-Luc Cote

Revised - May 8th, 2013

Cust. Repres.: Yutao Zhang

By - W.G. Crawford



Quality Control Record

Generator Heat Run

Q-1524-040

ITP Step # 1520-19

Project BC Hydro GMS

Unit : 4

Page 8 of 9

Generator Rise Summary

	Heat Run		
	1	2	3
Load MVA	18.29	70.93	83.67
Voltage L-L	13993.6	14456.0	14585.3
% V error	1.403	4.754	5.691
Air	27.77	28.07	28.57
Max RTD	55.77	84.60	95.23
PU MVA ²	0.033	0.503	0.700
Stator Rise	28.00	56.53	66.67
I ² R _f	144.18	380.06	467.87
Rise	22.0	49.6	58.2

Temperature Rise Rotor and Stator

Rotor Temperature Rise	
Rise - $A \times I^2R_f + B$	
A =	0.1130
B =	5.9337
R _f @ 90 C =	0.154384
I _f rated	1847.00
I ² R _{f 90(Kw)} =	526.67
Temperature = Rise + 40	
I ² R _f = I ² R _{f 90} x TCF	
TCF = $(234.5 + T) / (234.5 + 90)$	
solve for T =	108.9
Rise = T - 40 =	68.9
I ² R _f Actual	517.5

Note - Loss reference temperature Of 90 degree C was used

Stator Temperature Rise	
Stator Rise = $A \text{ pu MVA}^2 + B$	
A =	58.51
B =	26.28
Rise =	84.79

Measured by: Winston Crawford

Date :20/02/2013

ANDRITZ Rep. Jean-Luc Cote

Revised - May 8th, 2013

Cust. Repres.: Yutao Zhang

By - W.G. Crawford



Quality Control Record

Generator Heat Run

Q-1524-040

ITP Step # 1520-19

Project BC Hydro GMS

Unit : 4

Page 9 of 9

Instruments Used	Tool Number	Serial Number
Field Voltage Isolator	HCP 0255	11041969
Fluke 189 Digital Multimeter MH Tool	HCP 0180	78450040
Field Current	HCP 0184	84720075
Fluke 189 Digital Multimeter MH Tool		
Field Current Shunt - 7500 Amp - 100 MV		Customer's Device
Power	HCP 0035	12WC25606
Yokogawa WT2030 Digital Power Meter		
Temperature Instrument Customer PLC		---
Armature Voltage		Customer's
Potential Transformer - Ratio 14400/120		Device
Armature Current		Customer's
Current Transformer - Ratio 12000/5		Device

Measured by: Winston Crawford

ANDRITZ Rep.: Jean-Luc Cote

Cust. Repres.: Yutao Zhang

Date :20/02/2013

Revised - May 8th, 2013

By - W.G. Crawford